THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The roots of $x^2 - 8x + 15 = 0$ are: 1

(a) -3 and 5 **(b)** 3 and 5 **(c)** 3 and -5 **(d)** -3 and -5

2. Which of the following is irrational? 1

(a) $3 \times \sqrt{9}$ **(b)** $\sqrt{25} - 3$ **(c)** $\sqrt{9} + 3$ **(d)** $\sqrt{5} + 3$

3. If α, β are zeroes of $x^2 - 9x + 14$, then $\alpha\beta$ is: 1

(a) 15 **(b)** 14 **(c)** 9 **(d)** -14

4. The pair $x + 2y = 3$ and $2x + ky = 7$ has no solution when $k =$ 1

(a) 2 **(b)** 5 **(c)** 6 **(d)** 4

5. The discriminant of $2x^2 - 7x + 3 = 0$ is: 1

(a) 73 **(b)** 25 **(c)** -25 **(d)** 37

6. For which of the following is n such that 3^n ends with the digit 0? 1

(a) only even n **(b)** no natural number n

(c) $n = 2$ only **(d)** every n

7. The zeroes of $(x - 2)(x - 3)$ are: 1

(a) -2 and -3 **(b)** 2 and -3

(c) 2 and 3 **(d)** -2 and 3

8. The point where the lines $7x + 2y = -6$ and $3x + 5y = 14$ intersect has $x + y =$ 1

(a) 4 **(b)** -2 **(c)** -6 **(d)** 2

9. The nature of the roots of $2x^2 - 6x + 3 = 0$ is: 1

(a) three real roots **(b)** no real roots

(c) two equal real roots **(d)** two distinct real roots

10. The exponent of 2 in the prime factorisation of 480 is: 1

(a) 5 **(b)** 7 **(c)** 8 **(d)** 6

11. The sum of the zeroes of the cubic $x^3 - 7x^2 + 14x - 8$ is: 1

(a) 14 **(b)** 8 **(c)** 7 **(d)** -7

12. The pair of equations $2x + 3y = 8$ and $4x + 6y = 12$ is: 1

(a) exactly two solutions
(b) infinitely many solutions (coincident lines)
(c) exactly one solution (intersecting lines)
(d) no solution (parallel lines)

13. For the equation $x^2 - 9x + 20 = 0$, the sum of the roots is: 1

(a) 20 **(b)** -11 **(c)** 9 **(d)** -9

14. The exponent of 5 in the prime factorisation of 3600 is: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 2

15. The number of zeroes of a polynomial whose graph cuts the x-axis at 2 point(s) is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 1

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the sum and product of the zeroes of $2x^2 - 8x + 6$. 2

17. Solve by substitution: $x + y = 8$, $x - y = 2$. 2

18. Find the roots of $2x^2 - 7x + 6 = 0$ by factorisation. 2

19. Explain why $7 \times 6 + 7$ is a composite number. 2

20. Find a quadratic polynomial with sum of zeroes $\frac{1}{4}$ and product -1. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the roots of $x^2 - 6x + 8 = 0$ using the quadratic formula (if they exist). 3

22. Find the HCF and LCM of 26 and 91 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

23. Form a quadratic polynomial whose zeroes are 3 and 7. 3

24. The sum of the digits of a two-digit number is 9. If the digits are interchanged, the number becomes 45 less. Find the number. 3

OR

Find the values of k for which $x^2 + kx + 4 = 0$ has equal roots. 3

25. Prove that $\sqrt{3}$ is irrational. 3

OR

Find the zeroes of $2x^2 - 9x + 4$ and verify the relationship between the zeroes and the coefficients. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A sweet seller has 420 kg of kaju barfi and 130 kg of badam barfi. She packs them into trays of the same weight, each tray holding one kind only. Find the maximum weight of a tray and the number of trays. 5

OR

Given that 1 and 2 are two zeroes of $x^3 - 8x^2 + 17x - 10$, find the third zero. 5

27. The taxi charges consist of a fixed charge plus a charge per km. For 10 km the fare is ₹170 and for 15 km it is ₹240. Find the fixed charge, the charge per km and the fare for 25 km. **5**

OR

The length of a rectangular plot exceeds its breadth by 7 m and its area is 330 m². Find its dimensions. **5**

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND